

ROYAL FITNESS CLUB

Fitness Foundations Certificate

MODULE 4

Nutrition 101 — Fuelling Your Body

Duration	55 minutes
Course Code	CS_B1
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: Why Nutrition Is the Foundation

Food as Fuel, Structure, and Information

Nutrition is not merely about calories. The food you eat serves three fundamental roles: it provides energy (fuel) for all biological processes; it provides structural building blocks (protein, minerals) for tissue repair and growth; and it provides regulatory signals (vitamins, phytonutrients, omega-3s) that modulate gene expression, hormones, and inflammation.

The phrase "you cannot out-train a bad diet" is not cliché — it is physiology. A caloric deficit of 500 kcal per day takes 5 minutes to consume in a fast-food meal. Burning those 500 kcal through exercise takes 45–60 minutes of vigorous training. Nutrition provides the leverage.

Core Nutrition Principles

- Energy balance (calories in vs calories out) primarily determines body weight
- Macronutrient composition determines body composition (muscle vs fat)
- Micronutrients ensure metabolic processes function optimally
- Meal timing can modulate performance and recovery (though less important than total intake)
- Hydration affects performance, cognition, and every biochemical reaction in the body

SECTION 2: Macronutrients — The Big Three

Proteins — The Building Blocks

Protein is composed of amino acids — 20 in total, of which 9 are essential (cannot be synthesised by the body and must be obtained from food). Proteins function as structural components (muscle, collagen, hair, nails), enzymes, hormones (insulin, glucagon), immune factors (antibodies), and transporters (haemoglobin).

For fitness: Dietary protein provides the amino acids required for muscle protein synthesis (MPS). The most important amino acid for MPS is leucine — a branched-chain amino acid (BCAA) that directly activates the mTORC1 signalling cascade. A threshold of ~2–3g leucine per meal is required to maximally stimulate MPS.

Protein Quality Measure	Definition	High Scorers	Low Scorers
Biological Value (BV)	Ratio of nitrogen absorbed from protein to nitrogen in the diet	Eggs (100), Whey (104), Milk (91)	Wheat (54), Corn (60)

PDCAAS	Amino acid profile vs human needs	Casein (1.0), Soy (1.0), Whey (1.0), Wheat gluten (0.25)
DIAAS	Updated; digestion in small intestine	Beef (~1.1), Egg (~1.1) Most plant proteins <1.0

Protein Requirements for Different Goals

Population	Recommendation (g/kg BW/day)	Notes
Sedentary adult	0.8	Minimum; prevents deficiency
Recreational exerciser	1.2–1.6	Maintains lean mass
Resistance training (hypertrophy)	1.6–2.2	Upper range during caloric deficit
Endurance athlete	1.4–1.7	Protein for oxidation + repair
During caloric deficit (fat loss)	2.2–3.1	Prevents muscle loss (leucine-rich)
Older adults (>60)	1.6–2.2	Anabolic resistance requires more protein per meal

Protein distribution: Spreading protein intake across 3–4 meals (20–40g each) maximises 24-hour MPS compared to skewing protein to one meal. This is a practical reason for breakfast protein — skipping it reduces daily MPS potential.

Carbohydrates — The Performance Fuel

Carbohydrates are the body's preferred fuel for moderate-to-high intensity exercise. They are classified by structure:

Class	Chain Length	Examples	Digestion Speed
Monosaccharides	1 unit	Glucose, fructose, galactose	Immediate
Disaccharides	2 units	Sucrose (table sugar), lactose, maltose	Fast
Oligosaccharides	3–10 units	FOS (prebiotic fibres)	Slow/fermented
Polysaccharides	10+ units	Starch (amylose/amylopectin), glycogen	Slow → very slow
Dietary fibre	Non-digestible	Cellulose, pectin, beta-glucan	Not digested — fermented

Glycaemic Index (GI) and Glycaemic Load (GL)

Glycaemic Index rates how quickly a food raises blood glucose relative to pure glucose (GI=100). **Glycaemic Load** accounts for portion size: $GL = (GI \times \text{grams of carb}) / 100$. GL is more practically relevant — watermelon has a high GI but a low GL due to its high water content.

GI Category	GI Range	Examples	Sports Context
Low GI	<55	Oats, legumes, basmati rice, most fruits	Endurance (sustained energy)

Medium GI	55–69	White rice, corn, banana	Mixed — versatile
High GI	>70	White bread, rice cakes, sports drinks, dates (rapid glycogen resynthesis)	

Carbohydrate loading: In the 24–72 hours before endurance events lasting >90 minutes, increasing carbohydrate to 8–12g/kg/day maximises glycogen stores (~500g total: ~400g muscle + ~100g liver). This delays glycogen depletion and "hitting the wall" (bonking).

Fats — The Essential Macronutrient

Dietary fat is not the enemy — it is essential for hormone production (testosterone, oestrogen, cortisol all require cholesterol), fat-soluble vitamin absorption (A, D, E, K), cell membrane integrity, brain function (60% of the brain is fat), and as the primary fuel at rest and low intensities.

Fat Type	Structure	Examples	Health Effects
Saturated	No double bonds	Butter, coconut oil, tallow	Neutral at moderate intake; excess may increase LDL
Monounsaturated (MUFA)	1 double bond	Olive oil, avocado, nuts	Improves lipid profile; anti-inflammatory
Polyunsaturated (PUFA)	2+ double bonds	Omega-3 (fish, flaxseed, hemp), Omega-6 (sunflower)	Essential; improves lipid profile; anti-inflammatory; balance omega-6:3
Trans fats	Artificially hydrogenated	Partial hydrogenation products (margarine)	Increases CVD risk; minimise or avoid

Omega-3 fatty acids: EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid) are the most bioactive omega-3s. They reduce inflammation, improve insulin sensitivity, support brain health, and may enhance muscle protein synthesis. Target 1–3g EPA+DHA per day via fatty fish (2×/week) or fish/algae oil supplementation.

Dietary fat requirement for athletes: 20–35% of total calories from fat; never below 15% (disrupts hormone production). Fat intake below 15% suppresses testosterone production — counterproductive for any athletic goal.

SECTION 3: TDEE — Total Daily Energy Expenditure

Calculating Your Caloric Target

TDEE (Total Daily Energy Expenditure) is the total calories your body burns in 24 hours. It is composed of four components:

Component	Abbreviation	% of TDEE	Description
Basal Metabolic Rate	BMR	60–75%	Calories to maintain basic life functions at rest
Thermic Effect of Activity	TEA	15–30%	Calories burned during planned exercise

Non-Exercise Thermogenesis	NEAT	All the rest outside planned exercise (walking, fidgeting, posture, etc.)
	TEF	Calories burned digesting food (protein TEF highest ~25-30%)

Mifflin-St Jeor Equation (Most Accurate Widely Used)

BMR for males: (10 × weight kg) + (6.25 × height cm) – (5 × age) + 5

BMR for females: (10 × weight kg) + (6.25 × height cm) – (5 × age) – 161

Activity Level	Multiplier	Example
Sedentary (desk job, no exercise)	1.2	BMR × 1.2
Lightly active (1–3 days/week exercise)	1.375	BMR × 1.375
Moderately active (3–5 days/week)	1.55	BMR × 1.55
Very active (6–7 days hard exercise)	1.725	BMR × 1.725
Extremely active (physical job + training)	1.9	BMR × 1.9

Caloric Targets by Goal

Goal	Deficit/Surplus	Weekly Weight Change	Notes
Aggressive fat loss	-750 kcal/day (max)	~0.7kg/week loss	Risk of muscle loss above this
Moderate fat loss	-500 kcal/day	~0.5kg/week	Optimal for body composition
Gentle fat loss	-250 kcal/day	~0.25kg/week	Best for athletes maintaining performance
Maintenance	0	0kg	Recomp possible for beginners
Lean bulk	+200–300 kcal/day	~0.1–0.2kg/week	Minimises fat gain
Standard bulk	+500 kcal/day	~0.25–0.5kg/week	Faster mass, more fat gain

SECTION 4: Meal Timing for Performance

The Anabolic Window — Fact and Fiction

The "anabolic window" concept — that protein must be consumed within 30 minutes post-workout or gains are lost — is largely a myth when total daily protein intake is adequate. Research shows the window is likely 4–6 hours around training when factoring in pre-workout protein.

However, the peri-workout (around training) window does matter when:

- Training in a fasted state (morning without breakfast)
- Performing two training sessions in one day

- Training sessions exceed 90 minutes
- Competing in endurance events with rapid recovery requirements

Timing	Recommendation	Rationale
Pre-workout (1–3 hrs before)	20–40g protein + carbohydrate + vitamins	Amino acids available during training; glycogen topped up
Intra-workout (>60 min sessions)	30–60g carbs/hr (sports drink, banana, M&J's)	Maintains blood glucose; spares glycogen
Post-workout (within 0–2 hrs)	20–40g complete protein + carbohydrate	MPS initiation; glycogen resynthesis
Before sleep	30–40g casein, slow digesting	Slower digesting; sustains MPS overnight; GH release during sleep

SECTION 5: Hydration

Water — The Forgotten Nutrient

The human body is 60–70% water by mass. Even mild dehydration (1–2% body weight loss as sweat) demonstrably impairs endurance performance by 5–10%, cognitive function, and mood. At 5% dehydration, heat stroke risk rises sharply.

Daily fluid needs: Approximately 35–45ml per kg body weight per day for sedentary adults. Add 500–750ml per hour of moderate exercise; more in heat and humidity.

Indicator	Optimal	Mild Dehydration	Severe Dehydration
Urine colour	Pale yellow (lemonade)	Dark yellow (apple juice)	Dark amber / orange
Body weight change	<1% loss	1–3% loss	>3% loss
Thirst sensation	Absent or mild	Noticeable	Intense
Performance impact	Baseline	–5–10%	–20–30%+

Electrolytes: Sweat contains sodium, chloride, potassium, magnesium, and calcium. For sessions under 60 minutes, water suffices. For longer sessions in heat, sodium (the primary sweat electrolyte at 500–2000mg/hr) must be replaced to prevent hyponatraemia — dangerously low blood sodium from over-hydrating with pure water.

Hydration Strategy for Training

- Drink 500ml water 2 hours pre-workout; another 250ml 15–20 minutes before
- Drink 150–250ml every 15–20 minutes during training
- Weigh before and after — each 1kg lost ≈ 1L of fluid to replace
- Electrolyte drink for sessions >60 minutes; hot weather; heavy sweaters
- Coffee/tea count toward hydration — the mild diuretic effect is offset by the fluid volume

SECTION 6: Reading Nutrition Labels

Decoding the Nutrition Facts Panel

Understanding nutrition labels is a practical skill that empowers food choice. Indian packaged food labels (FSSAI regulations) include serving size, energy, protein, carbohydrate (total and sugars), fat (total, saturated, trans), dietary fibre, and sodium.

Key Label Elements

- **Serving size:** The reference amount for all values. Always check — a packet may contain 2–3 servings
- **Calories per serving:** Total energy. Remember 1 cal on label = 1 kcal in science
- **% Daily Value:** Based on a 2000 kcal diet — adjust for your TDEE
- **Added sugars vs total sugars:** Fruit has natural sugar (less concerning); added sugar should be minimised
- **Sodium:** Target <2300mg/day for general health; athletes may need more if sweat rate is high
- **Ingredient list:** Ingredients listed in descending order by weight — first ingredient is most abundant

5/20 rule: 5% Daily Value or less = low in that nutrient; 20%+ = high. Apply to nutrients you want more of (fibre, protein, vitamins) and those you want to limit (saturated fat, sodium, added sugar).

Module 4 Key Takeaways

- Macronutrients (protein, carbs, fat) provide energy and structural/regulatory functions
- Protein: 1.6–2.2g/kg/day for athletes; distribute across 3–4 meals with 20–40g per meal
- Carbohydrates are the primary fuel for moderate-to-high intensity exercise; choose GI appropriately
- Fats are essential — never go below 15% of calories; prioritise unsaturated and omega-3 sources
- TDEE = BMR × activity multiplier; set caloric target by adding/subtracting from maintenance
- Hydration: pale urine, 35–45ml/kg body weight/day baseline; add 500–750ml per training hour
- Read labels critically — serving size, added sugar, sodium, and ingredient order matter most

1. What is PDCAAS and why does it matter for plant-based eaters?
2. Calculate the TDEE for a 30-year-old male, 75kg, 175cm, who trains 4 days/week.
3. Explain glycogen depletion ("bonking") and how carbohydrate loading prevents it.
4. Why is dietary fat below 15% of calories harmful for athletes?
5. What is the difference between Glycaemic Index and Glycaemic Load? Give an example where GI is misleading.